



EDUCATION AND ARTIFICIAL INTELLIGENCE

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RESEARCH REPORT

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English & Communication III - IT139 in

BSc Degree in Information Technology

Sri Lanka International Buddhist Academy

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DECLARATION

I do hereby declare that the work reported in this project report was exclusively carried out by me under the supervision of It describes the results of my own independent work except where due references has been made in the text. No part of this project report has been submitted earlier or concurrently for the same or any other degree.

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Date

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Signature of the candidate

Certified by:

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Date

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Signature

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ABSTRACT

Artificial intelligence (AI) is developing and its application is spreading at an alarming rate, and AI has become part of our daily lives. As a matter of fact, AI has changed the way people learn. However, its adoption in the educational sector has been saddled with challenges and ethical issues. The purpose of this study is to analyze the opportunities, benefits, and challenges of AI in education. A review of available and relevant literature was done using the systematic review method to identify the current research focus and provide an in-depth understanding of AI technology in education for educators and future research directions. Moreover, the research explores the efficiency of automated grading systems powered by AI, streamlining assessment processes for educators while providing timely and constructive feedback to students. The integration of AI into educational tools is examined, elucidating how intelligent technologies enhance interactive learning experiences, fostering engagement and skill development. The study also considers potential challenges and ethical considerations surrounding the implementation of AI in education, ensuring a comprehensive analysis of its implications. Ultimately, this investigation contributes to the discourse on the evolving landscape of education in the digital age, highlighting the significant role that AI plays in shaping the future of learning.

01.INTRODUCTION

Education is undergoing a transformative shift with the integration of Artificial Intelligence (AI) technologies. Teachers were seen using computers in teaching, researching, and recording students' grades and in doing other things. Similarly, students, among other things, made use of computers in studying, researching, and solving problems. Also, computers have been used as an educational resource (analogous to a library or laboratory), as well as a means for maintaining databases of student information. . Artificial Intelligence is “*the science and engineering of making intelligent machines*” or “*a machine that behaves in a way that could be considered intelligent if it was a human being*” (Mccarthy, 2007). Studies mainly in the developed countries have concentrated on challenges in the disruption of AI in Education whiles, the opportunities and benefits of AI in education have received infinitesimal attention. This study is one of the few that provides an integrated overview of the opportunities, benefits, and challenges that AI adoption presents to the educational discipline.

Today, AI technology is revolutionizing schools and classrooms and making jobs much easier for educators. Over the years there has been a slow but steady revolution in education from using the blackboard to the whiteboard and now the use of projector screens in most educational institutions. Although there has not been a significant advancement in the use of AI in education from the 20th century to date (Dickson, 2017), research has shown steady improvement of AI in education through the use of different virtual assisted applications (Carlson, 2018).Studies (Johnson, 2019) have shown that AI has become part of our daily lives.

AI technology is used in automatic parking systems, smart sensors for taking photos as well as personalized assistance, not forgetting that it has also changed the way people learn. Currently, educational materials have become accessible through smart devices (Johnson, 2019). With the Application of AI in educational institutions, several administrative tasks have been automated and others are still being automated to give educators more time to spend on students. Prospective options for enhancing the efficacy and efficiency of educational institutions include personalized learning, automated grading, and AI-driven instructional tools. In order to solve issues, improve implementation techniques, and guarantee that AI in education is in line with pedagogical objectives while encouraging inclusion and ethical considerations, continued study is crucial as

the subject develops. In order to fully utilize AI's potential to influence education in the future, educators, academics, and technology developers must work together, as highlighted by the synthesis of recent literature.

02. METHODOLOGY

Creating and distributing a questionnaire to gather both quantitative and qualitative data was part of the process. Students have access to the online version of the questionnaire along with explicit instructions on how to fill it out. To promote candid and transparent responses, the Anonymity was intended for the questionnaire and guarantee privacy.

The questionnaire included Likert scale questions, which assessed students' opinions about an investigation on the role of AI in education, including personalized learning, automated grading, and the use of AI in educational tools.

Survey link:

<https://docs.google.com/forms/d/e/1FAIpQLSefgsk7Z5joTJoeqDMjdRZUG6R9OT8jvYLHWcgUCM6Xsik7iA/viewform?vc=0&c=0&w=1&flr=0&gxid=-8203364>



Education And Artificial Intelligence

B I U  

I am Shashini Ekanayaka , student from BSC IT Degree at SIBA Campus, doing a research of "An Investigation on the role of AI in education, including personalized learning, automated grading, and the use of AI in educational tools" for the module English & Communication.

I am collecting this information only for the above given purpose and won't be share with any third party.

Please send this questionnaire back within two days from today.

Thank you very much for agreeing to answer this questionnaire.

Figure 01

03. RESULTS & DISCUSSION

The questionnaire was administered to 31 Peoples', who High School Students, Bachelor's Degree Holders, Master's Degree Holders, Doctorates' from the university area and the results are presented below.

➤ Question 1 indicated:

What is your highest level of education?

The distribution of respondents' greatest degree of education, divided into four groups (high school, bachelor's, master's, and doctorate), is shown in the following chart. The percentage value assigned to each category denotes the percentage of respondents belonging to that particular education level. Based on the information responders, 29% have completed high school. 48.4% of those surveyed are bachelor's degree holders. A Master's degree has been obtained by 16.1% of the respondents. Six percent of those surveyed are doctorates. The educational attainment levels of the respondents are clearly displayed in this chart, which emphasizes the higher percentage of respondents with Bachelor's degrees, followed by High School education, and lower percentages with Master's or Doctorate degrees.

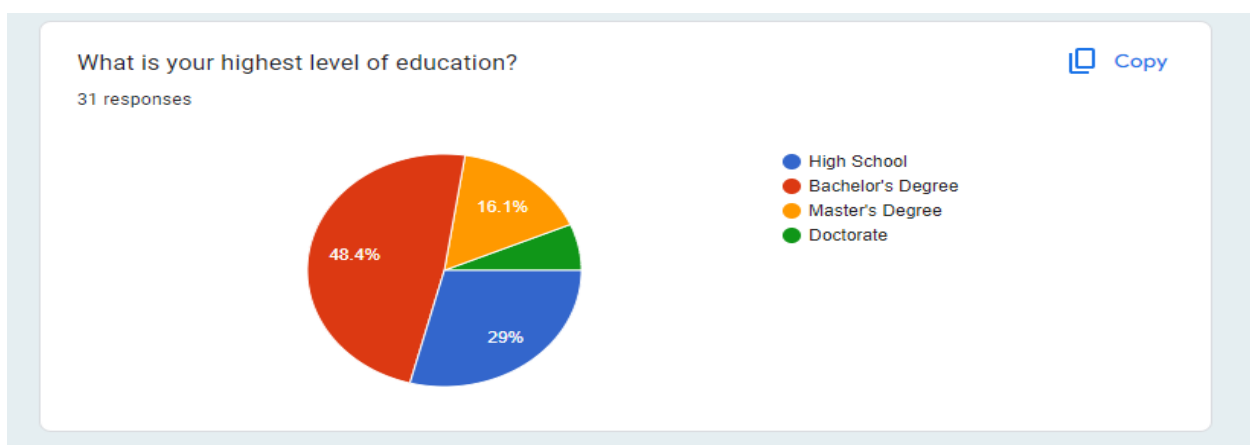


Figure 02

➤ Question 2 indicated:

Have you heard about the integration of Artificial Intelligence in education?

The responses of thirty-one people to the question, "Have you heard about the integration of artificial intelligence in education?" are displayed in a bar graph. Three categories are created from the responses: Yes, No, and Unsure. Based on the information, Indeed, according to 80.6% of respondents, artificial intelligence is being used in education. On the graph, this is shown by a comparatively tall bar that denotes a high proportion of responses. No, just 3.2% of respondents said they had never heard of artificial intelligence being used in the classroom. On the graph, this is shown by a noticeably shorter bar. 16.1% of respondents are unsure if they have heard about artificial intelligence being used in education. A bar of a moderate height on the graph also serves as a representation of this.

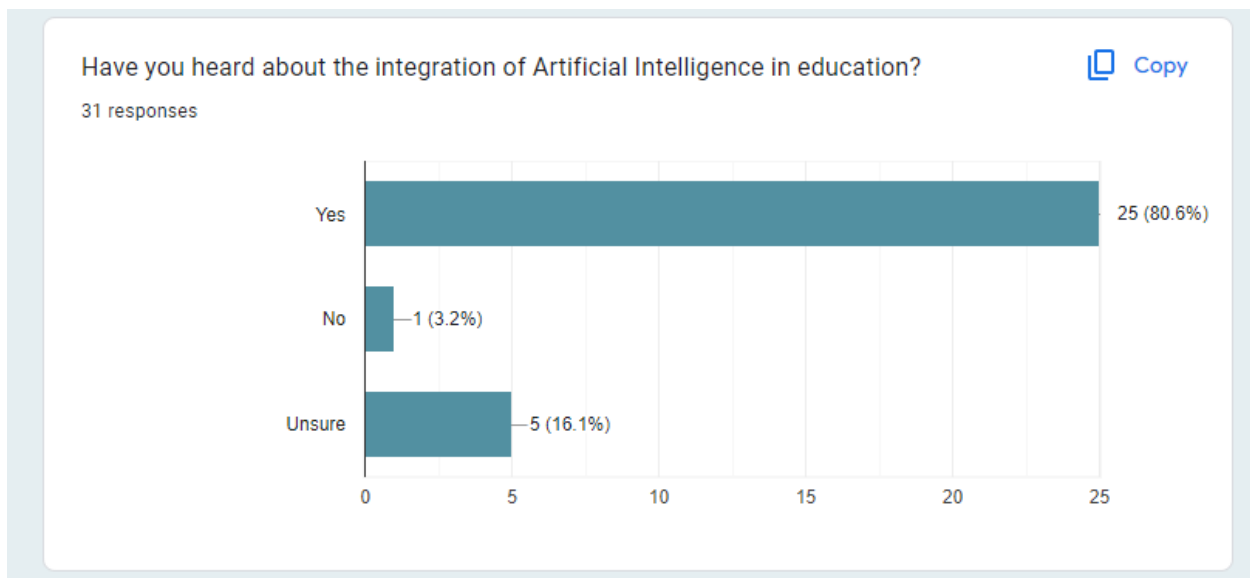


Figure 03

➤ Question 3 indicated:

Do you agree AI has the potential to positively impact education?

The pie chart represents the responses of 31 individuals to the question, "Do you agree AI has the potential to positively impact education?" The responses are categorized into four groups: Strongly Agree, Agree, Neutral, Disagree, and Strongly Disagree. According to the data, Strongly Agree, According to 32.3% of respondents, artificial intelligence (AI) has the potential to improve education. A section of the pie chart is used to represent this area. Agree, 61.3% of respondents, or the majority, concur that AI has the potential to improve education. The biggest area of the pie chart represents this. 6.5% of respondents are neutral regarding the possibility that artificial intelligence (AI) would improve education. A smaller piece of the pie chart is shown for this segment. Disagree and Strongly Disagree receive no responses, hence the pie chart does not show these categories.

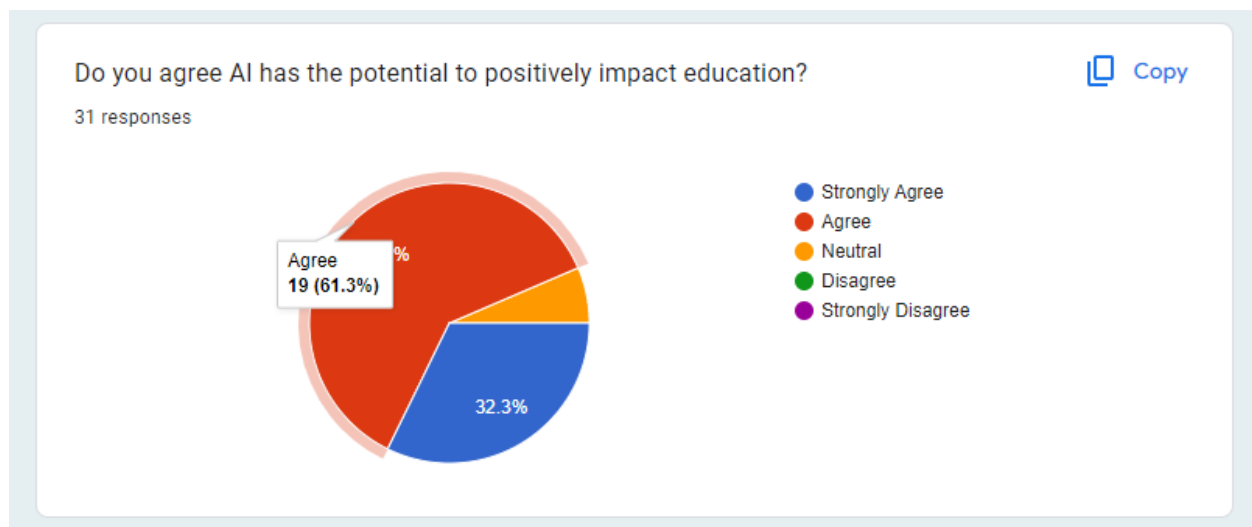


Figure 04

➤ Question 4 indicated:

Are you aware of any AI tools or technologies currently used in educational settings?

The pie chart represents the responses of 31 individuals to the question, "Are you aware of any AI tools or technologies currently used in educational settings?" The responses are categorized into four groups: Very aware, somewhat aware, Not very aware, and Not at all aware. According to the data, Very aware: According to 29% of respondents, AI tools and technologies that are now employed in educational settings are well known. A section of the pie chart is used to represent this area. Somewhat aware: 64.5% of respondents, or the majority, expressed some awareness of the AI tools and technologies that are now being utilized in educational contexts. The biggest area of the pie chart represents this. 3.2% of respondents said they were not very aware of the AI tools and technologies that are currently being employed in educational settings. A smaller piece of the pie chart is shown for this segment. Similar to the previous point, 3.2% of respondents had no idea what AI tools or technologies were being utilized in educational settings at all.

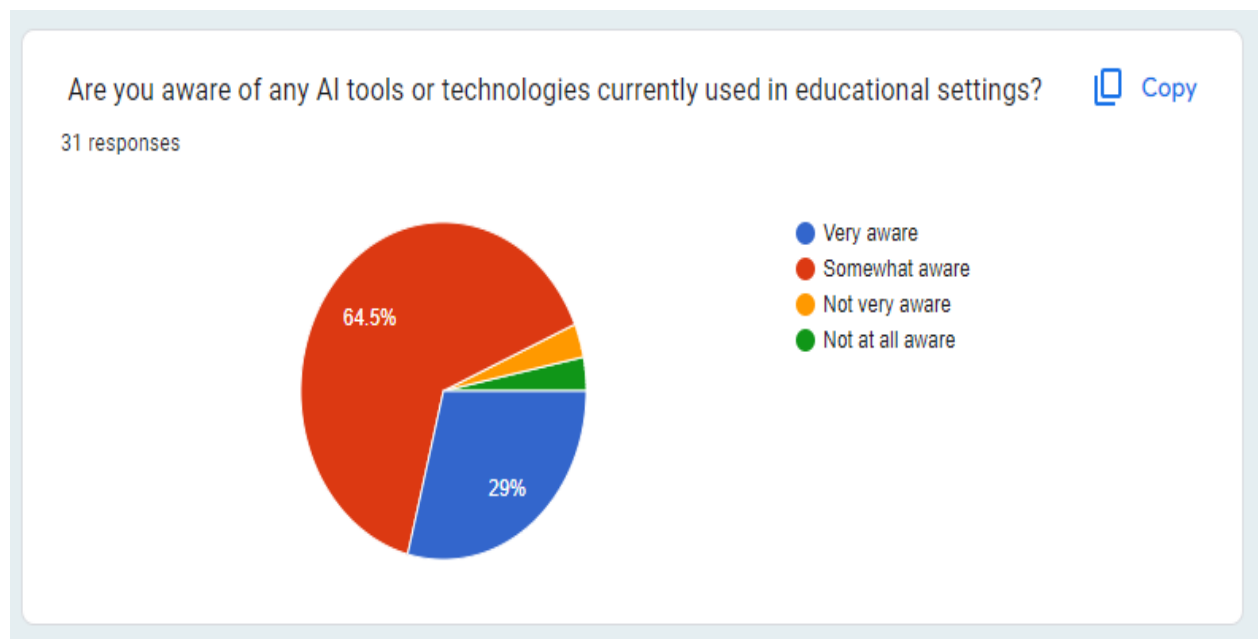


Figure 05

➤ Question 5 indicated:

Have you personally experienced the use of AI in your educational journey?

The pie chart represents the responses of the question, "Have you personally experienced the use of AI in your educational journey?" The responses are categorized into five groups: Always, Frequently, Occasionally, Rarely, and Never. According to the data, According to 32.3% of respondents, AI has always been used in educational processes. A section of the pie chart is used to represent this area. A further 32.3% of participants reported using AI in their educational process on a regular basis. The pie chart shows an equally significant chunk of this. According to 29% of respondents, AI has been used in their educational process on occasion. A slice of the pie chart that is somewhat smaller than the Always and Frequently categories is shown for this section. A lesser percentage of respondents, or 6.5%, said they had just seldom used AI in their educational process. A smaller piece of the pie chart is shown for this segment.

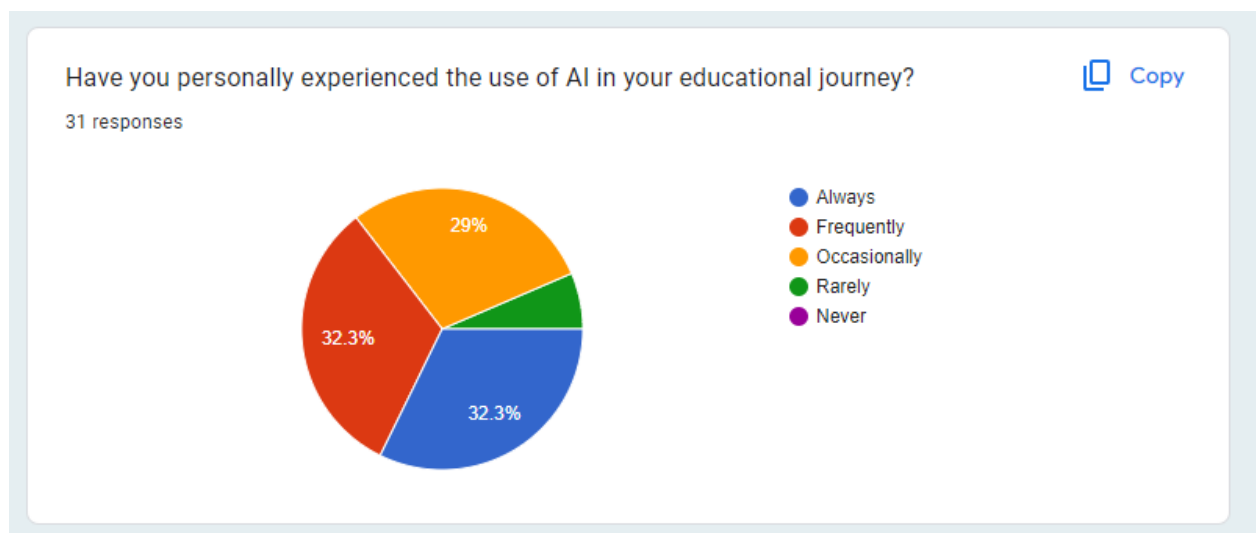


Figure 06

➤ Question 6 indicated:

In your opinion, how effective is AI in enhancing the learning experience?

The pie chart represents the responses of 31 individuals to the question, "In your opinion, how effective is AI in enhancing the learning experience?" The responses are categorized into four groups, Very Effective, Effective, Neutral, and Ineffective. According to 29% of respondents, artificial intelligence greatly improves the educational process. A section of the pie chart is used to represent this area. 48.4% of respondents, or the majority, think AI can effectively improve the learning process. The biggest area of the pie chart represents this. A significant percentage of participants, 19.4%, had no opinion about how well AI can improve the educational process. This area of the pie chart is represented as moderate. A tiny percentage of respondents 3.2% think AI has no influence on improving the educational process. A smaller piece of the pie chart is shown for this segment. Since there are no responses for Very Ineffective, this category does not appear in the pie chart.

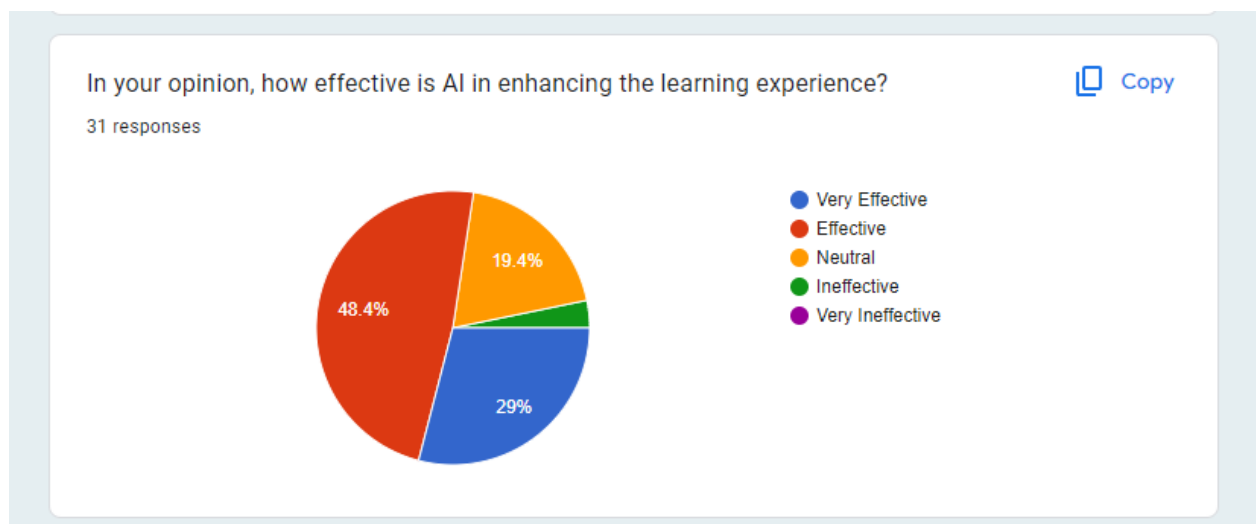


Figure 07

➤ Question 7 indicated:

Do you think AI can contribute to personalized learning experiences for students?

This chart represents the responses of the question, "Do you think AI can contribute to personalized learning experiences for students?" The responses are categorized into three groups: Yes, No, and Unsure. According to the data, Yes, according to the majority of responders (80.6%), AI can help provide students with more individualized learning experiences. The largest segment is shown in this area of the pie chart. No, according to a tiny percentage of responders (3.2%), AI cannot help provide pupils with individualized learning experiences. A smaller piece of the pie chart is shown for this segment. 16.1% of participants expressed uncertainty about AI's potential to enhance students' individualized learning experiences. This area of the pie chart is represented as moderate. Demonstrating the important part AI will play in determining how learning is done in the future.

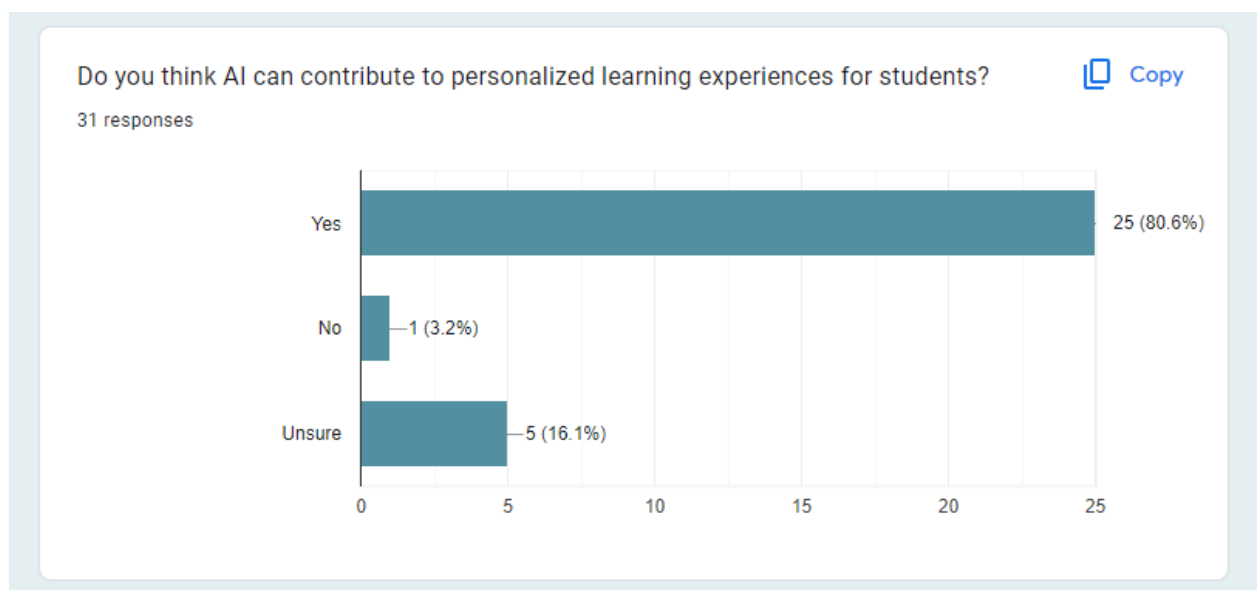


Figure 08

➤ Question 8 indicated:

Are you concerned about the ethical implications of using AI in education?

This pie chart shows the responses about "Are you concerned about the ethical implications of using AI in education?" The responses are categorized into five groups: Very concerned, Concerned, Neutral, Not concerned, and Not at all concerned. According to the data, extremely concerned: 16.1% of respondents expressed extreme concern over the moral ramifications of utilizing AI in the classroom. A section of the pie chart is used to represent this area. Concerned, of those surveyed, 45.2% expressed concern over the ethical ramifications of utilizing AI in the classroom. The biggest area of the pie chart represents this. Neutral, A significant percentage of respondents 25.8% stated that they have no concerns about the ethical ramifications of utilizing AI in education. This area of the pie chart is represented as moderate.

9.7% of respondents said they were unconcerned about the ethical ramifications of utilizing AI in the classroom. A smaller piece of the pie chart is shown for this segment. Not at all worried, 3.2% of respondents expressed no worry as all on the ethical ramifications of utilizing AI in the classroom. The pie chart shows a relatively minor piece of this region.

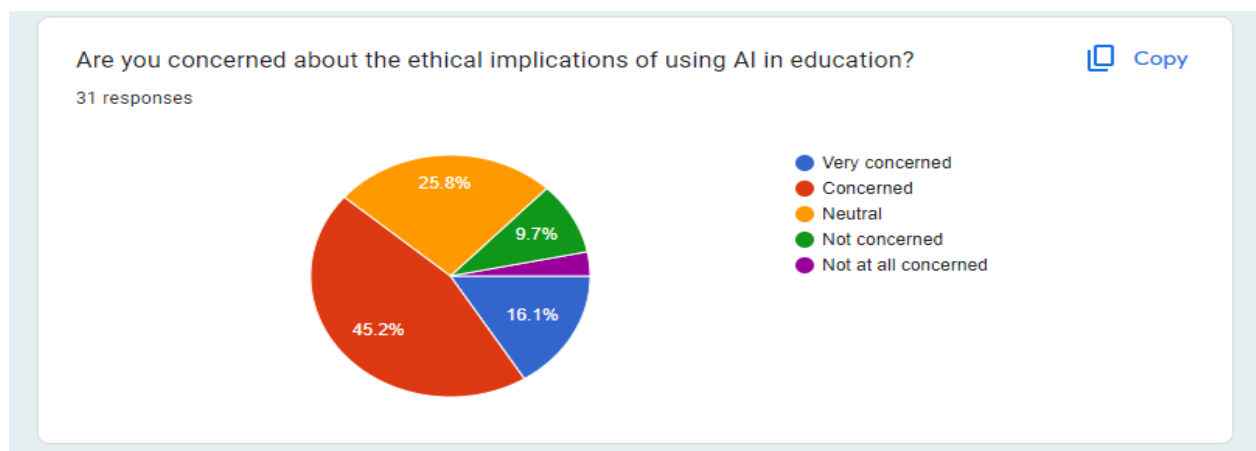


Figure 09

➤ Question 9 indicated:

Do you believe teachers should receive training on integrating AI into their teaching methods?

This bar chart represents the responses of 31 individuals to the question, "Do you believe teachers should receive training on integrating AI into their teaching methods?" The responses are categorized into three groups: Yes, No, and Unsure. According to the data, 74.2% of respondents, or most of the respondents, think that teachers ought to get training on incorporating AI into their lesson plans. On the chart, this is shown by a comparatively long bar, signifying a high proportion of responses. No, 9.7% of respondents, a lower percentage, disagree that educators should be trained in incorporating AI into their lesson plans. On the chart, this is shown by a noticeably shorter bar. 16.1% of respondents aren't sure if educators should be trained in incorporating artificial intelligence into their curricula. On the chart, a bar of moderate height also serves as a representation of this.

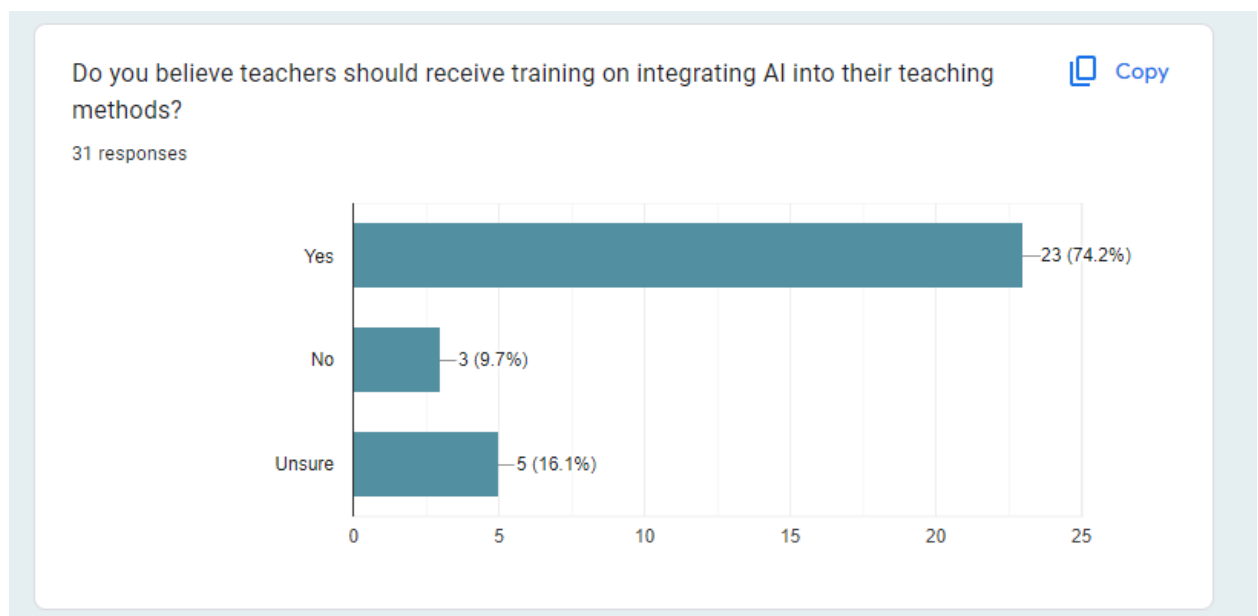


Figure 10

➤ Question 10 indicated:

What changes or developments do you expect to see regarding AI in education in the next five years?

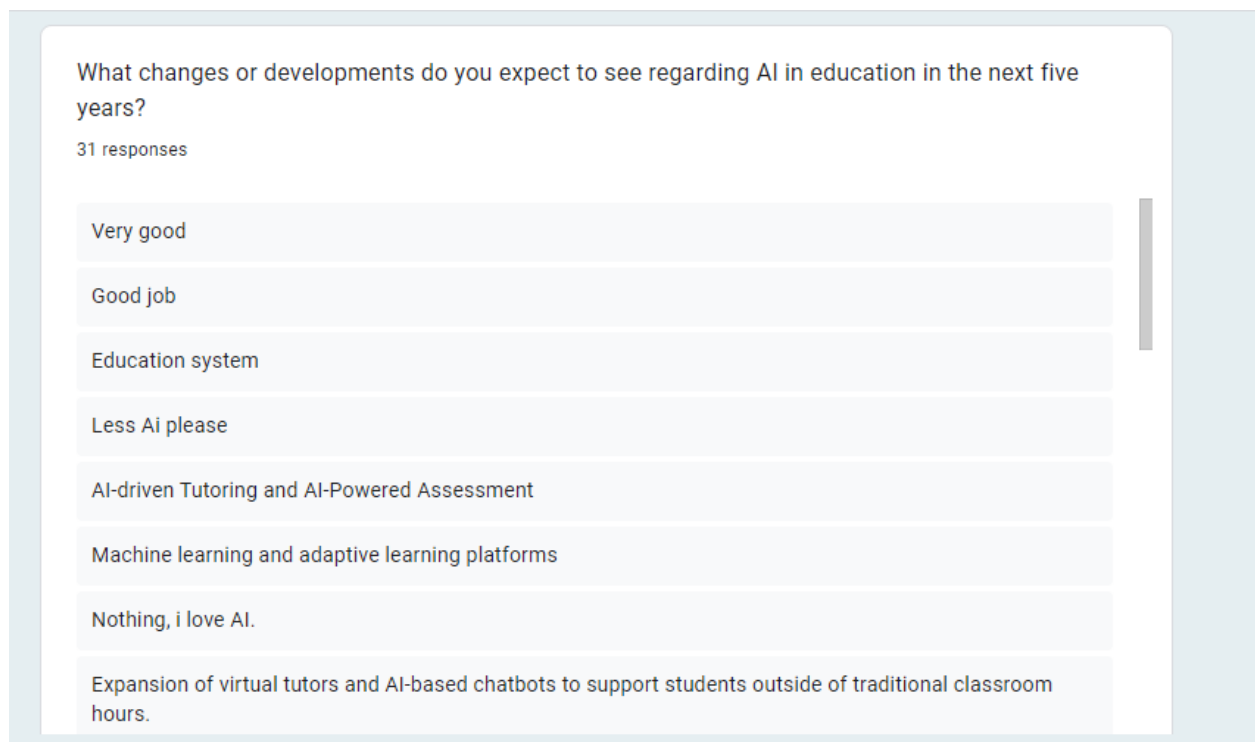


Figure 11

It should be more effective for students of all ages.
Modify AI tools for education systems
.
In the next five years, I expect to see AI in education advancing in personalized learning, adaptive assessment, virtual tutors, AI-driven content creation, and enhanced data analytics for student performance tracking.
AI subject must related to career feild
More privacy
AI-enabled technologies should positively integrate with current teaching and learning approaches.
Good

Figure 12

The main change should be made aware is limiting students from using generative tools such as chatGPT etc. In terms of development, a personalized AI tool which can progressively monitor students and acknowledge them for getting out of the track.
Nothing
In the next five years, AI in education is expected to bring about personalized learning experiences, adaptive learning platforms, virtual assistants for instant support, data-driven decision making, automated grading and assessment, and enhanced collaboration among students and teachers. These developments will revolutionize teaching and learning processes, improve student outcomes, and support educators in their roles.
Recommend AI as a school subject 8 - advance level
I think it's better to it guide to there own work not to al based AI work andalso AI is future of education of next Generation
Expansion of AI tutors and virtual assistants for individualized support.

Figure 13

04. LIMITATIONS

In this Research reports on education and artificial intelligence (AI) may face several limitations, including:

- There may not always be easy access to thorough and excellent data on student performance, educational procedures, and AI applications in learning environments.
- The use of AI in education brings up a number of ethical issues, including algorithmic bias, fairness, and data privacy.
- Due to differences in educational institutions, cultural contexts, and student demographics, research findings on AI in education may not be universally applicable.
- The process of education is intricate and multifaceted, impacted by a range of elements such as student backgrounds, socioeconomic status, curriculum design, and teaching approaches.

05. RECOMMENDATIONS

- To evaluate the long-term effects of AI initiatives in education, conduct longitudinal research.
- Examine the ethical issues related to using AI in education, such as algorithmic bias, data privacy, and the effects of AI-driven decision-making on educational equity. Provide strong policy frameworks that handle these issues and guarantee the ethical application of AI in learning environments.
- Address issues with sustainability and scalability when implementing AI solutions in a variety of educational contexts, such as settings with limited resources and impoverished areas. Examine cutting-edge strategies for growing AI-driven interventions while guaranteeing long-term sustainability and fair access.

06. CONCLUSION

In conclusion, this research has explored the role of AI in education. The knowledge obtained from the information gathered via a Google Form survey offers insightful viewpoints on this complex relationship. The use of AI in education, but the ones that explicitly discuss the relationship between opportunities, advantages, and difficulties are still few and dispersed. Some opportunities for students is by engaging them in collaborative learning and to the teachers by relieving them off extra duties such as administrative and marking of exams. This, in turn, enables teachers and educators to make time for research and increase contact hours with students who require that. Artificial intelligence technologies are an exciting area for human kind; however, as the participants in this research have implied, it is not a cure for everything or an improvement that will bring absolute good. Therefore, the legal, ethical, pedagogical, psychological and sociological harms and benefits are to be considered. Since it is humanity that is most affected by technology, it is important that this entire process is carried out on a legal basis, so as not to harm anyone.

REFERENCES

- Ahmet Gocen, Fatih Aydemir, (2020), Artificial Intelligence in Education and Schools : Research on Education and Media 12(1):13-21.
- K. Flamm, Creating the Computer: Government Industry and High Technology, Washington, DC, USA:Brookings Institution Press, 1988.
- M. Campbell-Kelly, Computer Student Economy Edition: A History of the Information Machine, Evanston, IL, USA:Routledge, 2018.
- M. M. L. Cairns, "Computers in education: The impact on schools and classrooms" in Life Schools Classrooms, Singapore:Springer, pp. 603-617, 2017.

APPENDIXES

➤ Google Form Link

<https://docs.google.com/forms/d/e/1FAIpQLSefgsk7Z5joTJoeqDMjdRZUG6R9OT8jvYLHWcgUCM6Xsik7iA/viewform?vc=0&c=0&w=1&flr=0&gxid=-8203364>